An internal row gives a nonterminal separator at odd degree

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le \mathrm{Alt}(\Omega)$ be transitive, with a connected string C-marking and all full maximal omissions intransitive. Assume $n = 2r + 1$ and an actual transitive six-point head supported on $D = \{1, \dots, 6\}$. Retain the literal hypotheses

$$\tau = \rho_3 = (6\ b)\delta, \quad \mathrm{supp}(\delta) \subseteq \Omega \setminus (D \cup \{b\}), \qquad R = G_{\ge 3} = \mathrm{Alt}(\Omega \setminus \{1, \dots, 5\}).$$

Put $\sigma = \rho_4$, $H = G_{\ge 5}$ and $K = G_{>5}$. Assume σ fixes D, while H fixes $D \cup \{b\}$. As in the acquired pure-overlap proof, $x = b^\sigma \ne b$, $\delta x = x$, and $\langle \tau, \sigma \rangle$ contains a pure three-cycle on $\{6, b, x\}$. These follow from the displayed actual natural R and full IP, without a rank induction.

Theorem 1. *Let O be an actual H-orbit of size $2q$, $q \ge 3$, which δ preserves and on which it acts nontrivially. Suppose K is intransitive on O. Let $F \subseteq \{6, \dots, r-1\}$ have size $q-1$, with an ambient-eligible edge of every colour in F contained in O. Then F is a consecutive interval, and its last label z satisfies*

$$6 \le z < r - 1, \qquad |\mathrm{Fix}(G_{>z})| \ge 2z + 3.$$

In particular it supplies the strict nonterminal separator needed at this internal-orbit equality.

Proof. The involution δ centralizes H, as H fixes $6, b$ and centralizes τ. It is free on the transitive orbit O. Moreover $\tau \in G_j$ for each $j \in F$, and its restriction to O is $\delta|_O$. The complete acquired internal-pair theorem therefore applies with $k = 5$. Write $a = \min F$ and $z = \max F$. It gives

$$F = \{a, \dots, a + q - 2\}, \qquad z = a + q - 2,$$

and $K|_O$ is the diagonal natural S_q on two rows. Every tail mark outside $F \cup \{5\}$ fixes O pointwise. The acquired prefix-row growth and its additional-fixed-point lemma, using the seven actual H-fixed points $D \cup \{b\}$, give the stronger estimate

$$|\mathrm{Fix}(G_{>z})| \ge (a-3)q + 8. \tag{1}$$

Both acquired lemmas hold before the last label as well as at it; they do not assume the critical even-degree equation. Writing $a = 6 + e$, $e \ge 0$, the difference between (1) and $2z + 3$ is

$$q - 3 + e(q-2) \ge 0.$$

This proves the required fixed-point bound.

Suppose that $z = r - 1$. Then every point is fixed by $G_{>z}$ and $n = 2r + 1 = 2z + 3$. Equality in the last inequality is necessary, so $q = 3$, $e = 0$, $a = 6$, $z = 7$, $r = 8$ and $n = 17$. The two $K = \langle \rho_6, \rho_7 \rangle$ rows of O have three points each. The complement of $D \cup \{b\} \cup O$ is a four-point set V, preserved by H and K. The point x lies in V: it is outside $D \cup \{b\}$ and is fixed by δ, which is free on O. Since K fixes b and commutes with σ, it fixes x as well.

The group $R = \langle \tau, \sigma, H \rangle$ is transitive on its natural support. Since τ and H preserve O, σ cannot preserve O. It centralizes K, so it must send one of the two three-point K-orbits in O to a three-point K-orbit outside O. All points of $D \cup \{b, x\}$ are fixed by K. Therefore that new orbit is exactly $W = V \setminus \{x\}$. The two involutions ρ_6, ρ_7 act transitively on W, and hence as two distinct transpositions generating natural S_3. They likewise act as the two path transpositions on each row of O. Their product has order three on all three rows and they fix the remaining points. Thus K itself is diagonal S_3 on these three natural rows.

Its centralizer on their union is exactly S_3 permuting the rows: each natural row has trivial centralizer and the equivariant bijection between any two rows is unique. The involution δ swaps the two rows of O, since it is free on O; it fixes W pointwise, since it preserves the three-point K-orbit W and centralizes its transitive action. It also fixes D, b, x. The involution σ swaps b, x, fixes D, and must exchange W with one of the two rows of O because it does not preserve O. Hence their exterior row actions are two distinct transpositions. On the nine-point row union the product $\tau\sigma = \delta\sigma$ has order three. On $\{6, b, x\}$ it is also a three-cycle, and it fixes the other five head points.

Thus $\langle \tau, \sigma \rangle$ is a dihedral group whose nonidentity rotations have nontrivial restrictions on both sets. It cannot contain the acquired pure three-cycle on $\{6, b, x\}$. This contradiction excludes $z = r - 1$ and finishes the proof. □

Attribution and boundary. The internal-pair equality, row-growth estimate and extra fixed point are complete acquired proofs. The new argument eliminates their only possible terminal equality at $n = 2r + 1$, using three actual natural S_3 rows and the pure-cycle intersection. No quotient IP or new computation is used. The existence of such a saturated nonfirst internal orbit remains a hypothesis; other orbit deficits and pairs split across different orbits are still needed for the full separator refinement. No whole-original, formal or question credit is claimed.